Supplementary methods for location grouped tactile evaluation

Note S1 Observation roles and preprocessing

The physical grouping unit is one registered location. Both loads and all lighting, noise, feature and model representations sharing that location remain in its group. Twelve locations produce 24 mechanical loading fields; lighting and noise do not increase that physical count.

The original and outer-refit procedures use the first 16 loaded noise frames for positive training. Their 64-frame no-touch threshold pool is divided into training indices 0–31 and calibration indices 32–63. Inner splits instead use no-touch training indices 0–15 and validation indices 16–31. A distinct 64-frame no-touch set is reserved for false-alarm evaluation. Outer loaded evaluation uses all 64 noise frames.

Let R be the 16-reference-frame mean in DN. The variance and three image channels are given below. No unregistered variance floor is added; clipping is applied to the standardized difference.

$$v=\left[max\left(R,0\right)\cdot 255/20000+\left(3\cdot 255/20000\right)^{2}+1/12\right]\cdot 17/16$$

$$D=clip\left(\frac{I-R}{\sqrt{v}},-10,10\right);\ X=\left[\frac{R}{255},D,\left|D\right|\right)$$

Native image network and objective

The four 3×3 convolutional layers are 3→12 (stride 1), 12→16 (stride 2), 16→24 (stride 2) and 24→16 (stride 1), all with padding 1 and sigmoid-weighted linear unit (SiLU) activation. A 1×1 convolution produces spatial logits h. Attention is softmax(h) over pixels; f is the weighted sum of 16 feature channels and q the weighted mean of normalized image coordinates. Strength u = logsumexp(h) − log(HW). Detection is Linear(17,1) on [f,u]; localization is Linear(18,32), Tanh, Linear(32,3) on [f,q].

Coordinates are normalized using means and standard deviations of training-location targets only, with each standard deviation at least 10−6 mm. Detection uses a batch mean of class-balanced BCE with class weights N/(2N+) and N/(2N0). Regression equals Σi yi ||p̂i − p̃i||² / [3m(N+/N)]. The denominator uses the dataset positive fraction; replacing it by the observed batch positive count changes the implemented objective.

The original run uses Adam at learning rate 0.001, batch size 8, gradient-norm clipping at 5, 64 epochs and seeds 1701, 1702 and 1703. Float32 and deterministic algorithms are enabled; automatic mixed precision and TensorFloat-32 are disabled. Adam is cited by its formal ICLR 2015 publication. Library defaults not explicitly passed to the historical optimizer are retained through the recorded environment rather than presented as newly tuned settings.

Algorithm S1 Location grouped nested selection

1. For each LED4 direction {0°,12°,34°,60°}, assign location coordinate index 0, 1 or 2 to its outer group; each group contains one location per region.

2. Hold out one outer group. Alternate the two remaining location groups as inner training and validation sets. Fit coordinate normalization using inner training targets.

3. Evaluate each learning rate {0.001,0.0003} and checkpoint budget {64,128,256} with seeds {1701,1702,1703}. Keep no-touch training, validation and threshold roles separate.

4. Average the inner validation objective over the two splits and three seeds. Select the minimum; break ties by lower epoch budget and then lower learning rate.

5. Reinitialize the chosen configuration for each seed and fit on the eight outer-training locations. Calibrate τ as the 0.95 quantile of the dedicated calibration scores.

6. Evaluate s > τ on independent no-touch data and all held-out loaded frames. Denormalize XYZ and save unconditional chord errors and joint success at radii 1,2,5,10 mm.

7. Save configuration, input identity, weights, optimizer and random-number state, loss histories, threshold and predictions. Compare paired output hashes before counting results as distinct.

Original 18 dimensional scalar features

At window scales k = 4, 8, 16, use reference gradients after Gaussian smoothing with σ = 0.6 pixel, standardized by √v. With box denoting the registered window reduction, define A = box(gx²)+ρ, B = box(gxgy), C = box(gy²)+ρ, bx = box(gxd), by = box(gyd), and ρ = 10−10[box(gx²)+box(gy²)]+10−12.

Define Q = (Cbx²−2Bbxby+ A by²)/(AC−B²) and E = box(d²−1)/√(2k²). Each scale contributes max E, max Q, mean of the largest 32 Q values, E at argmax Q, and the two components [(Cbx−Bby)/(AC−B²), (Aby−Bbx)/(AC−B²)] at argmax Q. Apply sign(x) log(1+|x|) to these 18 entries.

The extended 38-dimensional representation retains indices 0–17. Indices 18–21 hold two peak XY positions normalized by 1280 and 720, 22–23 window sizes divided by 16, 24–29 two nominal surface XYZ mappings divided by 100, 30–31 validity flags, and 32–37 nearest REST-marker anchor XYZ divided by 100. Invalid mappings use zero XYZ with their validity flag retained. The anchor is the nearest marker in image-plane Euclidean distance.

Multi marker representation and lightweight baselines

The multi-marker input uses 211 static markers at 9-, 17- and 25-pixel windows. Rounded marker centers follow rint(pixel−0.5) with a saved validity mask. Seven basis columns [gx,gy,u gx,v gy,−v gx+u gy,1/σ,Rsmooth/σ] are L2-normalized with floor 10−8. A Cholesky factor satisfies $LL^T = B^TB+10-6I$ and $z = L-1B^Td$. These seven coordinates are whitened projections, not seven directly recovered physical parameters.

Two more coordinates are $(d^Td-n)/\sqrt{\max(2n,1)}$ and $[d^Td-z^Tz-\max(n-7,0)]/\sqrt{\max(2(n-7),1)}$. Signed-log compression and scale → marker → coordinate ordering give 3×211×9 = 5697 entries.

Principal component analysis is fitted on training data only, retaining at most 12 components above max(10−12, λmax·10−12). Whitening uses singular values divided by √max(1,N−1), floored at 10−3. Input standard deviations are at least 10−6. Detection PCA includes positive and negative training data; localization PCA uses positives.

Ridge regression uses λ = 0.5 with an unpenalized bias. The lightweight multilayer perceptron (MLP) has one 24-unit Tanh hidden layer. Its base configuration uses full-batch Adam, learning rate 0.008, β1 = 0.9, β2 = 0.999, ε = 10−8 and weight-gradient decay 10−3 on weights only. Base training is 300 steps; historical 600/1200/2400/4800-step protocols preserve their own run identities. Initialization uses first-layer standard deviation 1/√d, output-layer standard deviation 0.15 and zero biases. Detection uses softmax cross-entropy and score output1−output0; thresholding is strict > at the calibration 0.95 quantile.

Data S1 Saved results and identity records

Supplementary Data S1 now includes both complete 288-condition CNN tables and a 72-row saved prediction-identity table, alongside the regional summary. Original and nested stages each cover 18,432 saved prediction rows. Twenty-one seed-paired prediction outputs are byte-identical. The platform additionally exposes the existing input/label families, local ADC inputs and feature-model diagnostics with their own source identities.

Implementation fingerprints: original CNN e4288040893eda3f2b7f712191bda15f0a18f615c68cdfb0af4434584e16e8aa; nested writer a96698456ef438f5832a1fc49fd7e34095f59abf8a6774e677bd916903321379; original 18-feature extractor 4d134afef08e3b855193b929f7b3920072b26941ac8a2184fc0c8720af1177ee; extended scalar source a0a7c879ee28898229a13a51c79723f116c79e03f5c81ce87c98a21433ecb1b2; multi-marker source 227a0044f598ef7cd80503dadee35526d3915e3a76f139c79b376987bfdff395; lightweight model source 45837bb37ea58c375f2aca0087b064075425899e2093572f4d51b39e842504e9.